# envBasta lahat ng ininstall natin for this project to work

# Installation Requirements and Dependencies

## 1. Overview

The ChainCert system is composed of three primary layers: the **frontend application**, the **backend API server**, and the **blockchain smart contract layer (chaincode)**. Each layer runs in a different runtime environment and therefore requires separate dependency installations.

The frontend runs on the Windows environment as a React application accessed through the user's browser. The backend API and blockchain components run inside the Ubuntu (WSL) environment because Hyperledger Fabric and its associated command-line tools require a Linux-based runtime. Supporting infrastructure components such as Docker and CouchDB are also required to operate the blockchain network and supporting services.

The following sections describe the required system tools, infrastructure components, and software dependencies necessary to deploy and run the ChainCert system.

---

# 2. System Requirements

## 2.1 Operating Environment

The development and deployment environment requires the following system-level components:

**Windows Environment**

- Windows 10 or Windows 11
- Windows Subsystem for Linux (WSL2)

- Docker Desktop (with WSL integration enabled)
- Git
- Node.js (version 18 or compatible)
- npm (Node Package Manager)

**Ubuntu Environment (WSL)**

- Ubuntu distribution installed through WSL
- Node.js (version 18 or compatible)
- npm
- curl
- wget
- Git

These tools support both the application runtime and the blockchain infrastructure required by the system.

---

# 3. Blockchain Infrastructure Dependencies

ChainCert uses **Hyperledger Fabric** as the underlying blockchain framework for credential issuance, verification, and audit logging. The following components must be installed and configured within the Ubuntu environment:

- Hyperledger Fabric binaries
- Fabric peer CLI tools
- Fabric Certificate Authority (Fabric CA)
- Docker and Docker Compose

- CouchDB containers for ledger state databases
- Fabric orderer service
- Fabric peer nodes for participating organizations

These components allow the system to create and manage blockchain networks, channels, and smart contracts used for credential operations.

---

# 4. Frontend Application Dependencies

The ChainCert frontend is implemented as a **React-based web application** and runs within the Windows environment. These dependencies must be installed within the frontend project directory.

### Core Frontend Libraries

- **react** – JavaScript library for building the user interface
- **react-dom** – DOM renderer used by React
- **react-router-dom** – Handles routing between application pages
- **react-scripts** – Build system and development scripts for React applications
- **axios** – HTTP client used for communication with the backend API

### User Interface Libraries

- **@fortawesome/fontawesome-free** – Provides icon assets used throughout the interface
- **web-vitals** – Performance monitoring library for measuring frontend performance metrics

### Credential Export Feature

- **jspdf** – Generates downloadable PDF files for issued credentials
- **qrcode** – Generates QR codes embedded within exported credential PDFs

### File Handling

- **multer** – Used for handling file uploads in certain frontend operations

### Testing Libraries (Optional)

The following libraries are used for testing and development validation but are not required for production deployment:

- @testing-library/react
- @testing-library/dom
- @testing-library/jest-dom
- @testing-library/user-event

### Frontend Installation Command

Run the following command inside the frontend project directory:

npm install react react-dom react-router-dom react-scripts axios jspdf qrcode @fortawesome/fontawesome-free multer web-vitals

Optional testing libraries may be installed using:

npm install @testing-library/react @testing-library/dom @testing-library/jest-dom @testing-library/user-event

---

# 5. Backend API Dependencies

The ChainCert backend server is implemented using **Node.js and Express** and runs within the Ubuntu WSL environment. The backend is responsible for managing authentication, interacting with CouchDB, handling issuer onboarding, and submitting transactions to the Hyperledger Fabric blockchain through the Fabric SDK.

## Core Backend Libraries

- **express** – Web framework used to implement the REST API
- **cors** – Enables cross-origin requests between the frontend and backend
- **dotenv** – Loads environment variables from configuration files

## Security and Authentication

- **bcryptjs** – Secure password hashing for user accounts
- **jsonwebtoken** – Implements JSON Web Token authentication for API access

## File Upload Handling

- **multer** – Handles multipart file uploads for issuer documents

## Notification Services

- **nodemailer** – Sends system notifications and email alerts

## Background Tasks

- **node-cron** – Schedules periodic background tasks such as credential expiration checks

## Docker Integration

- **dockerode** – Allows the backend to programmatically interact with Docker containers

## Blockchain Integration

- **fabric-network** – Hyperledger Fabric SDK used to submit transactions to the blockchain network
- **fabric-ca-client** – Used for enrolling identities through the Fabric Certificate Authority

### Backend Installation Command

Run the following commands in the backend directory:

```
cd ~/chaincert-api
npm install express cors dotenv bcryptjs jsonwebtoken multer nodemailer node-cron dockerode fabric-network fabric-ca-client
```

---

# 6. Chaincode (Smart Contract) Dependencies

ChainCert's credential issuance and verification logic is implemented as **Hyperledger Fabric chaincode**, which runs on the blockchain network. These dependencies must be installed in the chaincode directory within the Ubuntu environment.

### Chaincode Libraries

- **fabric-contract-api** – Framework for writing Hyperledger Fabric smart contracts
- **fabric-shim** – Provides the runtime interface between the chaincode and the Fabric peer

### Chaincode Installation Command

Run the following commands in the chaincode directory:

```
cd ~/chaincert-chaincode
npm install fabric-contract-api fabric-shim
```

---

# 7. Infrastructure Components

In addition to application dependencies, the system requires several infrastructure components to operate properly.

### Blockchain Network Components

- Hyperledger Fabric peer nodes

- Hyperledger Fabric ordering service
- Fabric Certificate Authorities
- Fabric channel configuration tools

### Database Infrastructure

- CouchDB containers used as the state database for Fabric peers
- CouchDB instances used for application data storage including:
    - user accounts
    - issuer records
    - credential metadata
    - notification records

### Containerization Environment

Docker is used to host Fabric peer nodes, certificate authorities, orderers, and CouchDB databases.

---

# 8. Dependency Management Note

Only **direct project dependencies** are documented in this section. Additional packages that appear within the `node_modules` directory are considered **sub-dependencies** automatically installed by npm as part of the dependency tree.

Examples of such sub-dependencies include:

- protobufjs
- uuid
- winston

- grpc-js
- nano

These packages are managed automatically by npm and do not require separate installation instructions.